

Rarefied hypersonic cylinder

Operator learning, baselines, and honest UQ

From author-supplied DSMC fields to a leakage-controlled teaching experiment

OUTCOMES

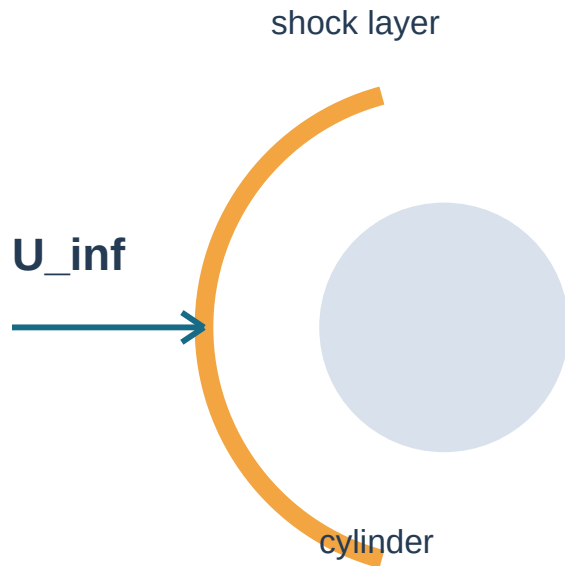
- Separate freestream Mach from the predicted local Mach field.
- Formulate a parameter-to-field operator for DSMC cylinder data.
- Freeze whole-case splits before model selection.
- Read the branch, trunk, layerwise fusion, and output head.
- Challenge the neural model with a structured interpolation baseline.
- Audit ensemble spread with empirical coverage, not visual confidence.

CLAIM

The lab executes a compact, author-released DSMC derivative. The default CPU model is a **teaching analog**; it does not reproduce the paper's full-resolution accuracy or speedup.

RULE

If a transparent baseline wins, retain that result. Complexity is not evidence.

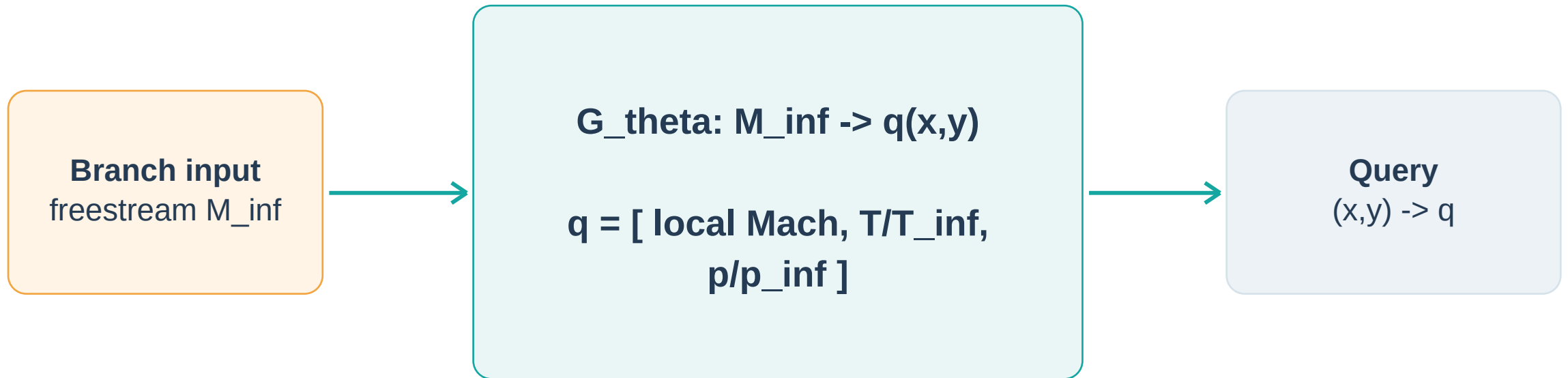


Mach number: $M_{\text{inf}} = U_{\text{inf}}/a_{\text{inf}}$ controls compressibility. The target $M(x,y)$ is a different, spatial field.

Knudsen number: $Kn = \lambda/D$ measures rarefaction. As λ becomes non-negligible, continuum closure and no-slip assumptions need qualification.

Sampling scale: DSMC fields estimate particle moments. A sharp shock and finite particle count make target noise spatially nonuniform.

The learning object is an operator



A field surrogate must be judged at unseen flow cases, not merely unseen grid points from a case already used during training.

Data audit: small derivative, complete provenance

20

Mach cases

400 x 400

source grid per case

50 x 50

deterministic selection

44,500

finite retained points

0.5 MB

committed NPZ

KEPT

- M_inf, x, y
- local Mach, T/T_inf, p/p_inf
- case ID and original source-grid row
- archive and derivative SHA-256 hashes

EXCLUDED

- 1.4 GB source ZIP and 4.9 GB expanded files
- training logs and duplicate checkpoints
- historical scripts with inconsistent protocols
- research checkpoints not needed for the lab

Whole-case splitting prevents a false claim

TRAIN

5, 6, 7, 8, 9, 10, 11, 12,
13, 14

VALIDATION

8.25, 8.75, 9.25, 9.75

BLIND INTERP.

5.5, 6.5, 7.5, 8.5, 9.5

BLIND EXTRA.

15

Wrong question: can the network fill random points from a Mach case it has already seen?

Scientific question: can it predict a complete field at an operating condition excluded from fitting and tuning?

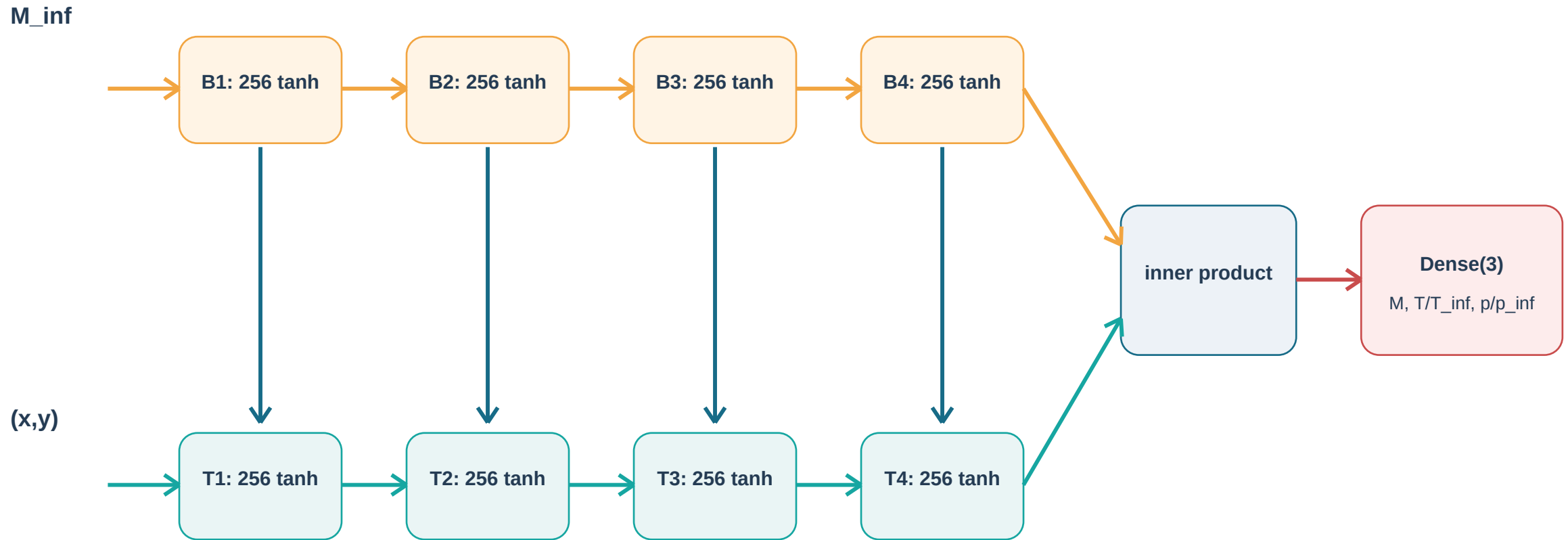
The mandatory baseline is field interpolation

$$\hat{q}(M_q) = q(M_a) + [(M_q - M_a)/(M_b - M_a)] [q(M_b) - q(M_a)]$$

Split	Local Mach	Temperature	Pressure
Blind interpolation	0.398%	0.609%	0.779%
Mach 15 extrapolation	0.538%	0.877%	0.848%

Result: a neural operator must beat a sub-1% structured baseline here, or justify itself through a different deployment constraint.

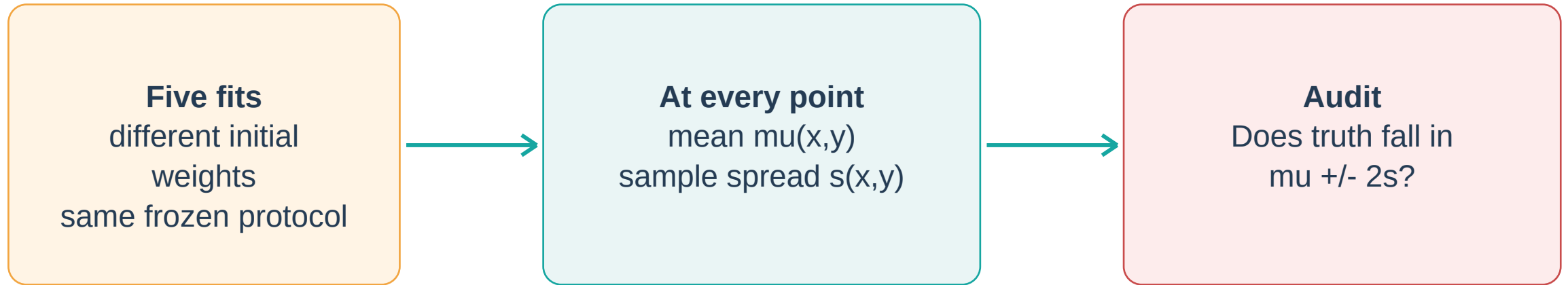
Reviewed Fusion-DeepONet topology



Each block includes dropout in the reviewed stored model. Branch state is added to the matching trunk depth before the final branch-trunk combination.

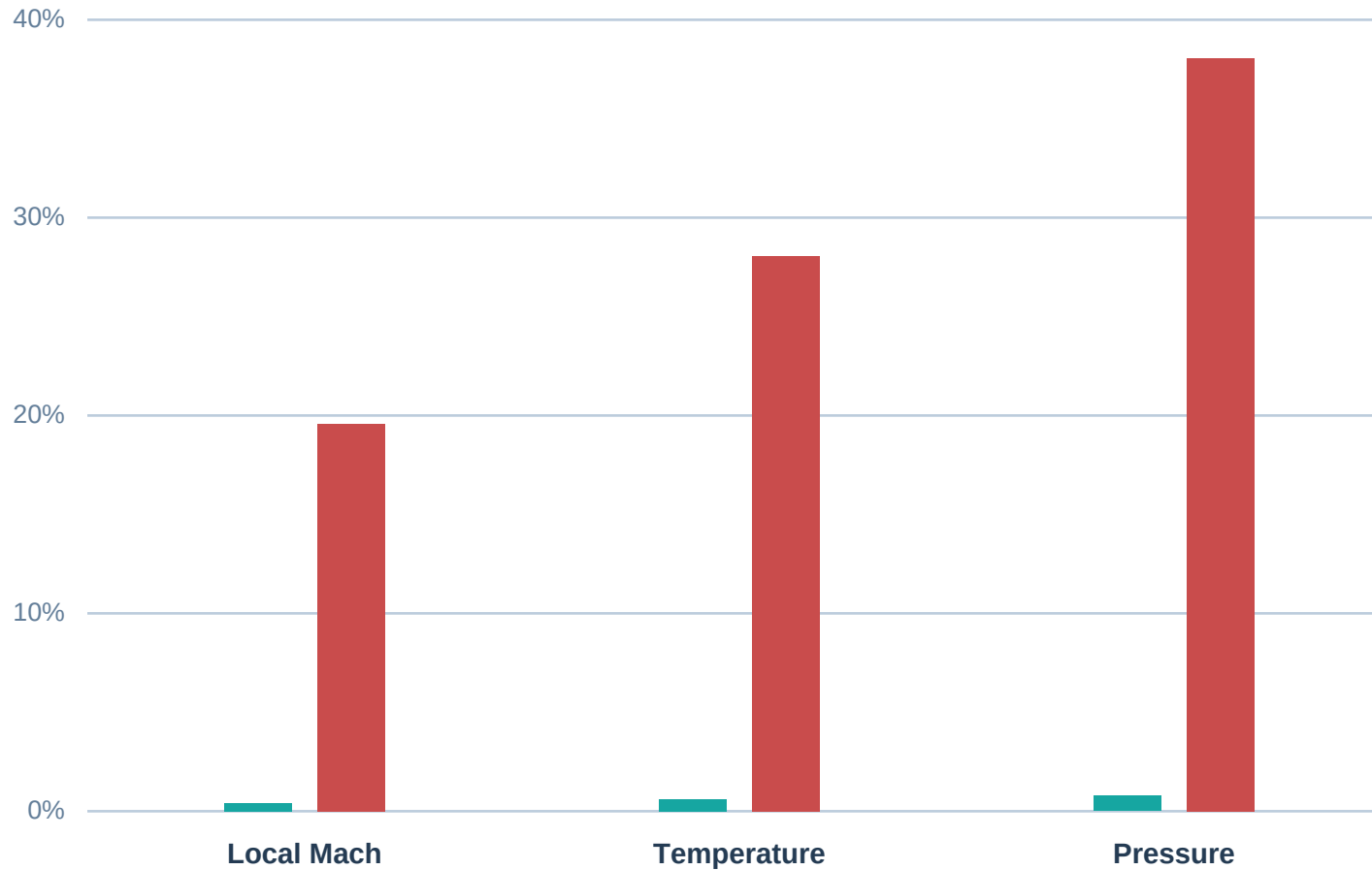
Training protocol: every choice changes the claim

- 1 Fit scalers** Training cases only; MinMax inputs and standardized targets.
- 2 Optimize** Adam with validation-monitored early stopping and learning-rate reduction.
- 3 Weight** Reviewed archive loss: $\text{MSE}_M + \text{MSE}_T + 5 \text{ MSE}_p$ in standardized space.
- 4 Repeat** Five independent initializations; preserve member predictions.
- 5 Open blind cases** After design is frozen; report interpolation and extrapolation separately.



Low coverage means overconfidence. High coverage can also be uninformative if intervals are too wide. Report coverage and interval width, by target and by split.

Executed evidence: the simple baseline wins

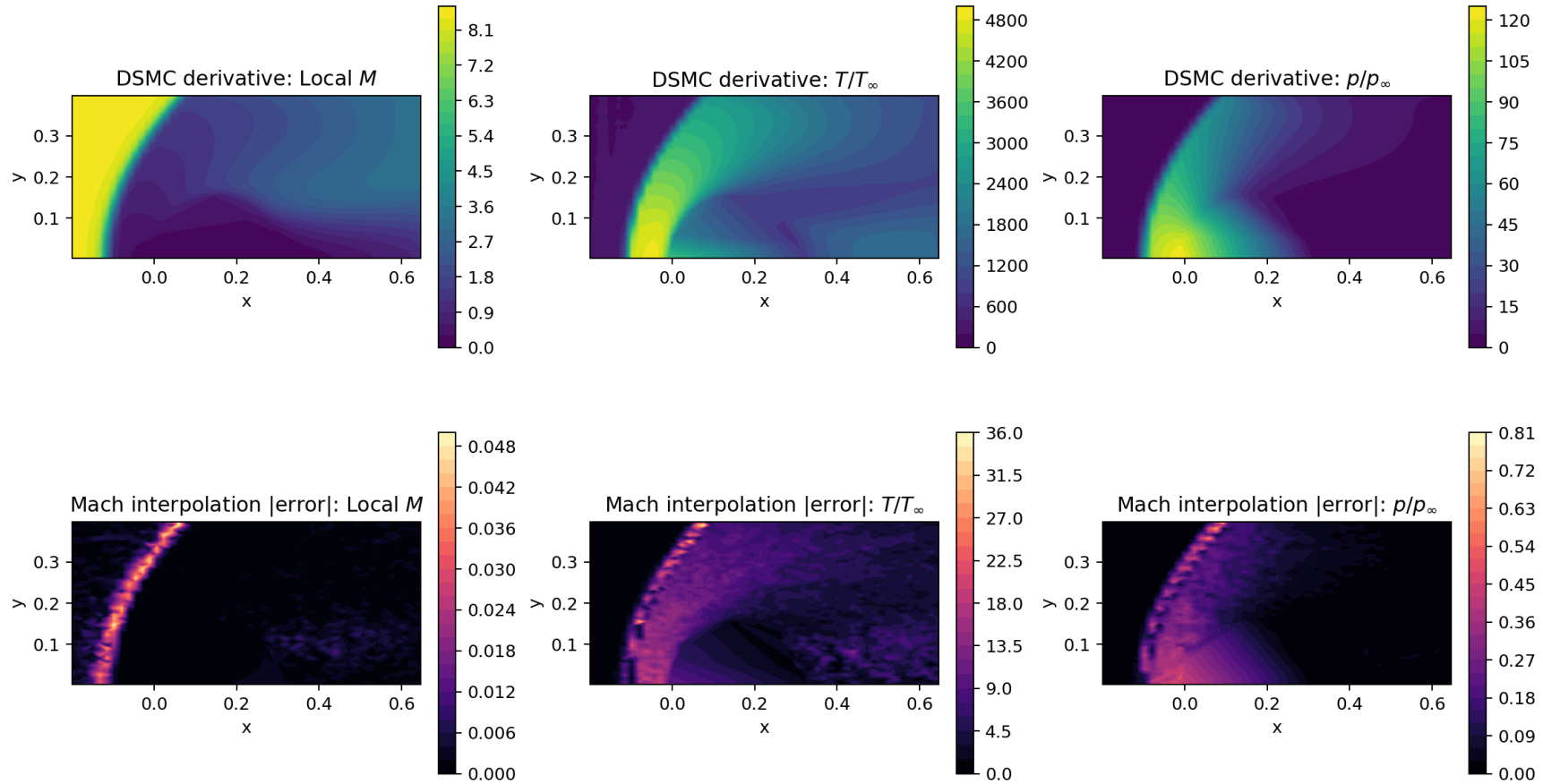


Baseline
0.398% | 0.609% | 0.779%

Teaching analog
19.6% | 28.1% | 38.0%

Where does interpolation error live?

Week 7.1 retained interpolation case: $M_\infty = 8.5$



Decision gate: when is a neural operator justified?

Accuracy

Does it beat field interpolation on frozen blind cases?

Geometry

Must one model span grids, shapes, or boundary conditions?

Availability

Will deployment lack bracketing high-fidelity solutions?

Cost

Do many field queries amortize training and data production?

Physics

Are shock location, surface loads, positivity, and limits acceptable?

Uncertainty

Is spread calibrated on the deployment regime?

SUBMIT

- Baseline and teaching-operator relative-L2 errors by target and split.
- One physics-based interpretation of an error hot spot.
- Two-sigma coverage and a calibration judgment.
- A short explanation of random-point leakage.
- One deployment condition that could justify full neural training.

MAY CLAIM

The compact DSMC derivative, frozen split, baseline, and CPU teaching analog were executed.

MAY NOT CLAIM

Published full-model accuracy, speedup, complete-field reproduction, or calibrated UQ.

- Roohi et al. (2026), Neural Networks for Rarefied Gas Dynamics: Relaxation Problem, Polyatomic Shock Waves, and Hypersonic Cylinder Flow. Physics of Fluids 38, 057108. <https://doi.org/10.1063/5.0334590>
- Lu et al. (2021), Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. Nature Machine Intelligence 3, 218-229. <https://doi.org/10.1038/s42256-021-00302-5>
- Lakshminarayanan, Pritzel, and Blundell (2017), Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. NeurIPS 30.
- Bird (1994), Molecular Gas Dynamics and the Direct Simulation of Gas Flows. Oxford University Press.

Rebuild data: [qa/build_hypersonic_cylinder_subset.py](#) | Rebuild evidence: [qa/run_hypersonic_cylinder_evidence.py](#) | Source archive SHA-256 begins bda221759a37